CS5234 Quiz 1 · Mock Paper

Paper · 试卷

Format (as announced): 50 minutes · open book · Lectures 1–5 · 4 MCQ (1 pt each) + 2 essay questions (9 + 7 pts) · a blank essay question earns 1 mark.

格式（按老师公布的要求）：50 分钟 · 开卷 · Lecture 1–5 · 4 道选择题（每题 1 分）+ 2 道大题（9 + 7 分）· 大题留空得 1 分。

Part I · Multiple choice (4 × 1 pt)

MCQ 1 (L1)

Let $X_1, \dots, X_k$ be pairwise independent Bernoulli(p) variables and $X = \sum_i X_i$. Which statement is always true?

(A) Var$(X) = kp(1-p)$

(B) Pr$(|X - kp| \geq \delta kp) \leq 2e^{-\delta^2 kp/3}$ for $0 < \delta \leq 1$

(C) $\mathbb{E}[X^2] = k^2p^2$

(D) X is independent of X_1

设 $X_1, \dots, X_k$ 两两独立，均服从 Bernoulli(p)，$X = \sum_i X_i$。下列哪项一定成立？

(A) Var$(X) = kp(1-p)$

(B) 当 $0 < \delta \leq 1$ 时 Pr$(|X - kp| \geq \delta kp) \leq 2e^{-\delta^2 kp/3}$

(C) $\mathbb{E}[X^2] = k^2p^2$

(D) X 与 X_1 独立

MCQ 2 (L2)

An estimator is within εA of the true value A with probability 0.6. How many independent repetitions are needed so that the median is within εA with probability $1 - \delta$?

(A) $\Theta(1/\delta)$

(B) $\Theta(\log(1/\delta))$

(C) impossible, because $0.6 < 2/3$

(D) $\Theta(1/\delta^2)$

某估计量以 0.6 的概率落在真值 A 的 εA 范围内。要使独立重复后的中位数以 $1 - \delta$ 的概率落在 εA 范围内，需要多少次重复？

(A) $\Theta(1/\delta)$

(B) $\Theta(\log(1/\delta))$

(C) 做不到，因为 $0.6 < 2/3$

(D) $\Theta(1/\delta^2)$

MCQ 3 (L3)

Which statement about Yao' s principle is correct?

(A) $D_\mu(f) \geq R(f)$ for every input distribution μ

(B) To prove $R(f) \geq q$ it suffices to exhibit one distribution μ with $D_\mu(f) \geq q$

(C) It only applies to problems with a 0/1 answer

(D) The hard distribution must be uniform over all inputs

关于 Yao 原理，下列哪项正确？

(A) 对任意输入分布 μ，$D_\mu(f) \geq R(f)$

(B) 要证 $R(f) \geq q$，只需找到一个分布 μ 使 $D_\mu(f) \geq q$

(C) 它只适用于答案为 0/1 的问题

(D) 难分布必须是全体输入上的均匀分布

MCQ 4 (L4)

For testing whether an array of n distinct numbers is sorted (ε-far means more than εn entries must change), which tester is correct with $O(\log n/\varepsilon)$ queries?

(A) Pick $O(1/\varepsilon)$ random i and check $a_i \le a_{i+1}$

(B) Pick $O(1/\varepsilon)$ random indices, sort them, and check the values are increasing

(C) Pick $O(1/\varepsilon)$ random indices and run a binary search for each value

(D) None: any tester needs $\Omega(n)$ queries

测试 n 个互异数组成的数组是否有序（ε-far 指需要改动超过 εn 项），下列哪个测试器在 $O(\log n/\varepsilon)$ 次查询下是正确的?

(A) 随机选 $O(1/\varepsilon)$ 个 i，检查 $a_i \le a_{i+1}$

(B) 随机选 $O(1/\varepsilon)$ 个下标，按下标排序后检查取值递增

(C) 随机选 $O(1/\varepsilon)$ 个下标，对每个值做一次二分搜索

(D) 都不行：任何测试器都需要 $\Omega(n)$ 次查询

Part II · Essay questions

Essay 1 (9 pts) · Counting equal pairs

You have query access to an array $a_1, \dots, a_n$. Let $N = \binom{n}{2}$ and let P be the number of pairs $i < j$ with $a_i = a_j$.
可查询数组 $a_1, \dots, a_n$。令 $N = \binom{n}{2}$，P 为满足 $a_i = a_j$ 的对 $i < j$ 的个数。

(a) [2] Using two queries, give an unbiased estimator of P.
用两次查询给出 P 的无偏估计。

(b) [3] Estimate P within additive error εN with probability $1 - \delta$ using $O(\varepsilon^{-2} \log(1/\delta))$ queries.
用 $O(\varepsilon^{-2} \log(1/\delta))$ 次查询，以 $1 - \delta$ 的概率把 P 估计到加性误差 εN 以内。

(c) [4] Now suppose we must decide exactly whether $P = 0$ or $P \ge 1$, with success probability $\ge 2/3$. Prove that $\Omega(n)$ queries are necessary. Hint: let μ_0 be $a_i = i$ for all i; for μ_1, copy a_1 into one random position.
现在要精确判断 $P = 0$ 还是 $P \ge 1$，成功率 $\ge 2/3$。证明需要 $\Omega(n)$ 次查询。提示：令 μ_0 为对所有 i 有 $a_i = i$；μ_1 把 a_1 复制到一个随机位置。

Essay 2 (7 pts) · Adjacent repeats in a stream

A stream $x_1, \dots, x_n \in [m]$ arrives one element at a time; n is not known in advance. Let $R = |\{i \le n-1 : x_i = x_{i+1}\}|$ and $\rho = R/(n-1)$. We want $\hat{\rho}$ with $|\hat{\rho} - \rho| \le \varepsilon$ with probability at least $2/3$.
数据流 $x_1, \dots, x_n \in [m]$ 逐个到达，n 事先未知。令 $R = |\{i \le n-1 : x_i = x_{i+1}\}|$，$\rho = R/(n-1)$。要以至少 $2/3$ 的概率输出满足 $|\hat{\rho} - \rho| \le \varepsilon$ 的 $\hat{\rho}$。

(a) [3] Design a streaming algorithm. Hint: sample a uniformly random position $I \in [n-1]$ with reservoir sampling, but you also need x_{I+1}.
设计一个流式算法。提示：用蓄水池抽样选一个均匀随机的位置 $I \in [n-1]$，但你还需要 x_{I+1}。

(b) [2] How large must k be? Justify.
k 需要多大？给出理由。

(c) [1] What is the space complexity?
空间复杂度是多少?

(d) [1] Why does it not work to simply store the first k adjacent pairs of the stream?
为什么不能简单地存下流中前 k 个相邻对?